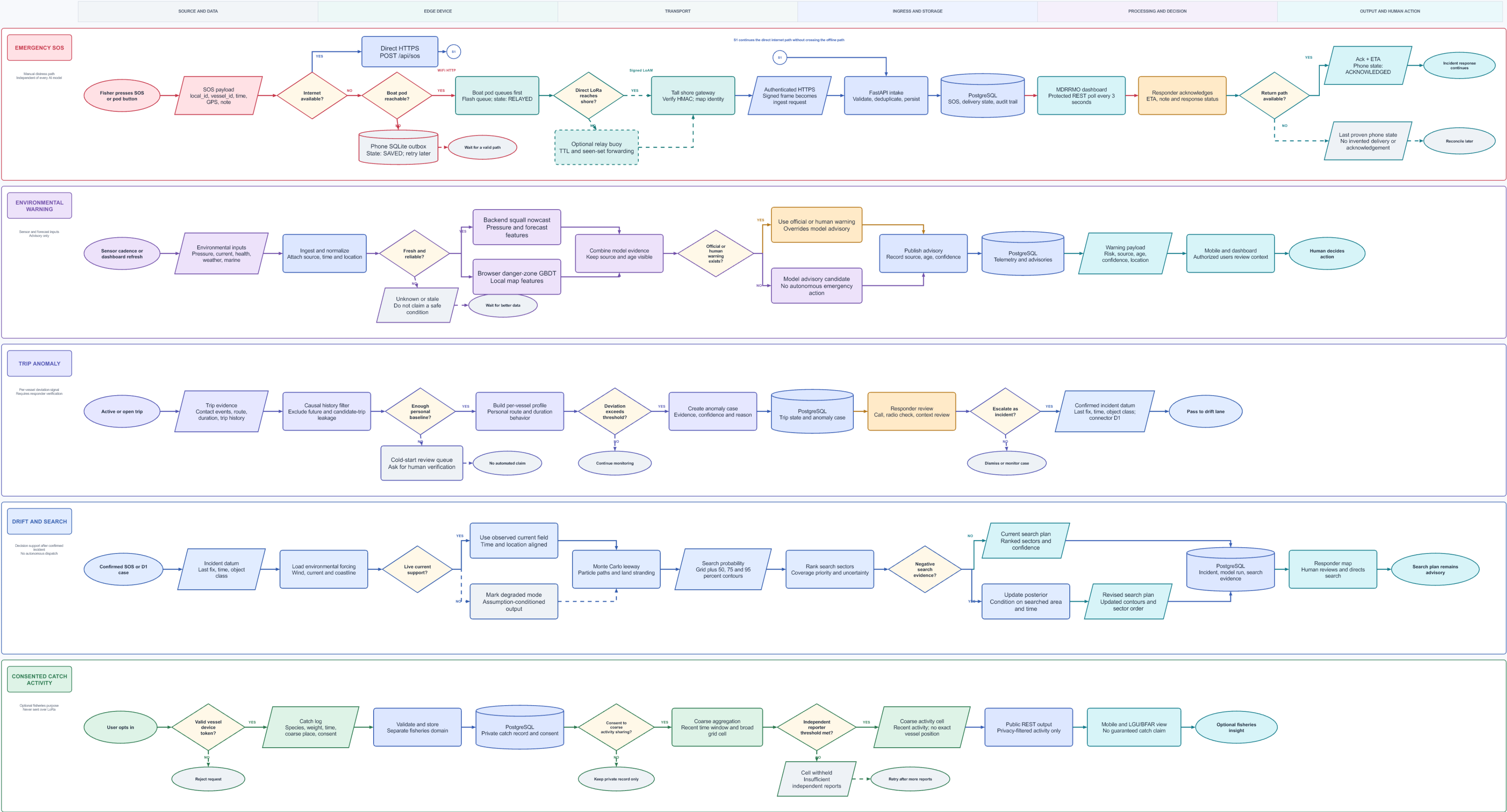


AqOne Aggregated Technical Architecture

Offline SOS transport, safety intelligence, storage, privacy controls and human response

CURRENT IMPLEMENTATION
Software paths tested locally; public deployment and field validation remain pending



ENVIRONMENTAL WARNING

Sensor and forecast inputs
Advisory only

Sensor cadence or dashboard refresh

Environmental inputs
Pressure, current, health, weather, marine

Ingest and normalize
Attach source, time and location

Fresh and reliable?

Backend squall nowcast
Pressure and forecast features

Browser danger-zone GBDT
Local map features

Combine model evidence
Keep source and age visible

Official or human warning exists?

Use official or human warning
Overrides model advisory

Publish advisory
Record source, age, confidence

PostgreSQL
Telemetry and advisories

Warning payload
Risk, source, age, confidence, location

Mobile and dashboard
Authorized users review context

Human decides action

Unknown or stale
Do not claim a safe condition

Wait for better data

Model advisory candidate
No autonomous emergency action

TRIP ANOMALY

Per-vessel deviation signal
Requires responder verification

Active or open trip

Trip evidence
Contact events, route, duration, trip history

Causal history filter
Exclude future and candidate-trip leakage

Enough personal baseline?

Build per-vessel profile
Personal route and duration behavior

Deviation exceeds threshold?

Create anomaly case
Evidence, confidence and reason

PostgreSQL
Trip state and anomaly case

Responder review
Call, radio check, context review

Escalate as incident?

Confirmed incident datum
Last fix, time, object class; connector D1

Pass to drift lane

Cold-start review queue
Ask for human verification

No automated claim

Continue monitoring

Dismiss or monitor case

DRIFT AND SEARCH

Decision support after confirmed incident
No autonomous dispatch

Confirmed SOS or D1 case

Incident datum
Last fix, time, object class

Load environmental forcing
Wind, current and coastline

Live current support?

Use observed current field
Time and location aligned

Monte Carlo leeway
Particle paths and land stranding

Search probability
Grid plus 50, 75 and 95 percent contours

Rank search sectors
Coverage priority and uncertainty

Negative search evidence?

Current search plan
Ranked sectors and confidence

Update posterior
Condition on searched area and time

Revised search plan
Updated contours and sector order

PostgreSQL
Incident, model run, search evidence

Responder map
Human reviews and directs search

Search plan remains advisory

Mark degraded mode
Assumption-conditioned output

CONSENTED CATCH ACTIVITY

Optional fisheries purpose
Never sent over LoRa

User opts in

Valid vessel device token?

Catch log
Species, weight, time, coarse place, consent

Validate and store
Separate fisheries domain

PostgreSQL
Private catch record and consent

Consent to coarse activity sharing?

Coarse aggregation
Recent time window and broad grid cell

Independent reporter threshold met?

Coarse activity cell
Recent activity; no exact vessel position

Public REST output
Privacy-filtered activity only

Mobile and LGU/BFAR view
No guaranteed catch claim

Optional fisheries insight

Reject request

Keep private record only

Cell withheld
Insufficient independent reports

Retry after more reports

FLOWCHART LEGEND



Solid: active software path Dashed: optional, unavailable or field-dependent path

NON-NEGOTIABLE ARCHITECTURE RULES
Manual SOS never depends on AI. Delivery states reflect evidence. Models advise authorized humans. Catch sharing stays optional and coarse.